

MOHD. TABREZ ASLAM MUKADAM

Systems & eBPF Security Researcher · Offensive Security (VAPT) · Adversarial AI Runtime

Email: tabrezmukadam57@gmail.com | LinkedIn: [linkedin.com/in/tabrez-mukadam](https://www.linkedin.com/in/tabrez-mukadam) | Portfolio: www.tabrez.tech | Location: Mumbai, India

PROFILE

Final-year BE student specializing in **Linux kernel & eBPF runtime security, offensive security/VAPT**, and **adversarial AI agent confinement**. Author of peer-reviewed IoT security research (ICSSSD 2026); disclosed **30+ validated vulnerabilities** across 15+ web & Web3 platforms; architected **Reality Kernel** (<0.31ms fast-path eBPF/LSM security barrier for AI agents); ranked **top 10% globally** (154 / 1,539 teams) in Snyk Fetch the Flag CTF. Technical author with 200K+ readers across InfoSec Writeups and CoinsBench.

TECHNICAL SKILLS

Systems & Kernel Security: eBPF / LSM Hooks (file_open, sys_enter_execve, bprm_check_security), Linux Internals, Syscall Tracing, Aya, Cgroup/PID Confinement

Offensive Security & VAPT: Web/API Pentesting, Threat Modeling, Exploit Analysis, Bug Bounty, Reverse Engineering, Purple Teaming, SOC & Incident Response

Adversarial AI Security: AI Agent Runtime Confinement, Prompt Injection / Jailbreak Mitigation, Insecure Tool-Calling Sinks, Ed25519 Cryptographic Receipts

Languages & Core: Rust, C / eBPF, Python, Bash, JavaScript, Solidity, SQL, Linux / Unix Internals, Git, Docker

Tools & Platforms: Burp Suite, Metasploit, Nmap, Wireshark, Ghidra, IDA Pro, OWASP ZAP, Remix, Hardhat, Microsoft Sentinel, HackerOne, Bugcrowd

FEATURED PROJECTS

Reality Kernel — eBPF Runtime Security Barrier for AI Agents Rust, Linux eBPF / LSM, Aya, Python, Ed25519

- Architected an in-kernel eBPF sensor attaching to LSM hooks (file_open, sys_enter_execve) to intercept agent syscalls and enforce fine-grained process confinement in kernel space before execution.
- Engineered a 4-layer Rust verification engine achieving <0.31ms fast-path latency with superposition intent mapping to block indirect prompt injections and unauthorized tool executions fail-closed.
- Implemented Ed25519 cryptographic receipts with SHA-256 state chaining (prev_hash) for tamper-evident audit trails; published client library [realitykernel](#) to PyPI.

RoninScan — On-Chain Threat Intelligence & Vulnerability Engine Node.js, Express.js, Etherscan API, Static Analysis

- Developed a static-analysis engine scanning smart contract bytecode and source for selfdestruct abuse, delegatecall hijacking, and reentrancy vectors; automated generation of client-ready PDF audit reports.

Protocol Zero & NutriPath AI — Web3 Security Curriculum & Agentic Privacy Vault Python, Gemini API, Solidity, EVM

- Authored open-source 10-chapter curriculum bridging Web2 VAPT to Web3 auditing. Built local differential-privacy vault isolating raw health telemetry from external LLM APIs (<15ms latency).

EXPERIENCE

Cybersecurity AI/ML Researcher (Intern) — SwiftSafe March 2026 – July 2026 | Remote

- Led adversarial-ML threat research under CTO supervision, mapping attack surfaces across LLM-integrated SaaS and autonomous agent pipelines.
- Built Python/Bash static-analysis tooling detecting prompt injection sinks, unsafe tool invocations, and unsanitized model I/O, reducing triage overhead.
- Authored enterprise-grade VAPT reports for AI deployments; evaluated jailbreak vectors, agent-tool data exfiltration, and blackbox adversarial fuzzing.

Cybersecurity Intern (Ethical Hacking) — Anonymous India Cyber Defence Academy Feb 2026 – March 2026 | Remote

- Delivered 12 end-to-end penetration tests from scoping to post-exploitation using Nmap, Metasploit, and custom Bash recon (SQLi, XSS, AV evasion).

Independent Security Researcher & Technical Writer — HackerOne, Bugcrowd, InfoSec Writeups June 2020 – Present | Remote

- Disclosed 30+ validated vulnerabilities (CORS, IDOR, XSS, access control) across web platforms. Audited 1,000+ lines of Solidity for DeFi protocols.
- Published 15+ technical deep-dives on systems security, AI runtime risk, and smart contract exploits, surpassing 200,000+ technical readers.

PUBLICATION

"When Smart Systems Optimize the Wrong Thing: Evaluating Security and Sustainability in a Tourist Incident Response System" ICSSSD 2026 | DOI: [10.64189/css.26407](https://doi.org/10.64189/css.26407)

- Designed a 3-layer adaptive IoT security framework (500 trajectories, 2,143 alerts); adaptive geofencing reduced energy use by 60.4%, while UI friction model lowered false escalations by 39.6%.

EDUCATION

B.E., Computer Science & Engineering (IoT, Cybersecurity & Blockchain) 2023 – 2027 (Expected) | CGPA: 7.70

Anjuman-I-Islam's M.H. Saboo Siddik College of Engineering, University of Mumbai

HONORS & CERTIFICATIONS

Honors & CTF: Rank #2, Blackbox 01 Web VAPT · Top 10% Global, Snyk Fetch the Flag CTF (154 / 1,539 teams) · Top 15% Rank, Hack The Box & TryHackMe · Smart India Hackathon Qualified (2024 & 2025, top 500+ teams) · Global Pathways AI Hackathon Outstanding MVP (Feb 2026)

Certifications: KC7 Advanced Persistent Analyst (2026) · DevTown x Microsoft Bounty Hunt 101 (2026) · arcX Threat Intelligence Analyst (2025) · EC-Council Android Bug Bounty Hunting (2025) · CyberWarFare Labs Purple Teaming (2024) · Cybrary Autopsy & Digital Forensics (2025) · Secure Google Cloud Network (2024)